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Feature Fusion: Enhancing Middle Ear Infection Diagnosis with Vision Transformer and Symptom Encoding

B Ananda Deva
Department of Computer Science and Engineering
Amrita School of Computing
Amritapuri, India
anandadeva2253@gmail.com

Varnika Sood
Department of Computer Science and Engineering
Amrita School of Computing
Amritapuri, India
varnikasood2004@gmail.com

A Fahim
Department of Computer Science and Engineering
Amrita School of Computing
Amritapuri, India
fahiiifahim@gmail.com

P M Arjun Krishna
Department of Computer Science and Engineering
Amrita School of Computing
Amritapuri, India
arjunkrishnapm777@gmail.com

Anjali T
Department of Computer Science and Engineering
Amrita School of Computing
Amritapuri, India
anjalit@am.amrita.edu

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***Abstract*—Acute otitis media and chronic suppurative otitis media are two examples of middle ear infections that are exceptionally hard to diagnose because of their overlapping symptoms and the subjective nature of standard otoscopy. This research presents a novel framework driven by Vision Transformer (ViT) to produce a cohesive diagnostic tool that addresses complexity in the real world. Utilizing the Kaggle "Otoscope Data" dataset, the model uses feature fusion, taking both the otoscopic image and clinical symptoms into account, with sophisticated preprocessing to uncover subtle patterns that traditional approaches often overlook. The system overcomes the constraints of sparse annotations and image variability through rigorous augmentation and adaptive learning, providing a solid approach that gives doctors an AI-driven solution using transformer model. This method helps us increase the accuracy of the clinical diagnosis, and also helps us to overcome the major challenges for underprivileged section of the society by enabling a cost-effective access to healthcare.**

***Index Terms*—Vision Transformer, Otoscopy Imaging, Symptom Encoding, Feature Fusion, Middle Ear Disease**

I. Introduction

Common middle ear infections that occur in a human body are Acute Otitis Media (AOM), Chronic Suppurative Otitis Media (CSOM), Tympanosclerosis, and Otitis Externa. Misdiagnosis or undiagnosed conditions can cause adverse side effects such as hearing loss or perforation of the tympanic membrane, even life-threatening diseases such as meningitis from such infections. So much of the conventional diagnostic approach relies on subjective otoscopic examinations, which are prone to variability, as the nuances of clinician experience and interpretation come into play. The shortage of qualified experts compounds the problem in rural or resource-limited environments, leading to late or incorrect diagnoses. This highlights the need for an automated, reliable, and readily available diagnostic system to assist healthcare workers in identifying middle ear infections accurately and efficiently.

There has been considerable interest in using artificial intelligence (AI) algorithms to automate otoscopy image analysis, particularly using convolutional neural network (CNN)-based approaches, with emerging success in the field of deep learning. However, there are inherent limitations in CNN-based methods that pose challenges for practical deployment [1]. First, CNNs are designed for local feature extraction which can fail to capture more global patterns and finer differences between infection types. Second, these models are frequently trained on small, imbalanced datasets with insufficient diversity, such as diversity of patient demographics, imaging conditions, or disease stages that often result in inferior generalizability. Third, most existing methods do not use clinical symptom data, which is an important part of diagnostic decision making in clinical practice. To address these limitations, there is a growing need for a more comprehensive and context-sensitive method of middle ear infection classification in order to better support clinical practice.

To address the above mentioned challenges, a novel framework based on vision transformers is presented that combines otoscopic image representations with encoding of the symptoms for improved diagnostic accuracy. Unlike CNNs, ViTs process images as sequences of patches, enabling the model to capture both local details and global contextual relationships within the otoscopic images. This feature is highly beneficial for differentiating between infections with similar appearances like the AOM and CSOM. Moreover, in the framework, patient-reported symptoms like pain, discharge, and hearing loss are included as extra input data. This symptom-aware approach localizes the model's reasoning to individual symptoms

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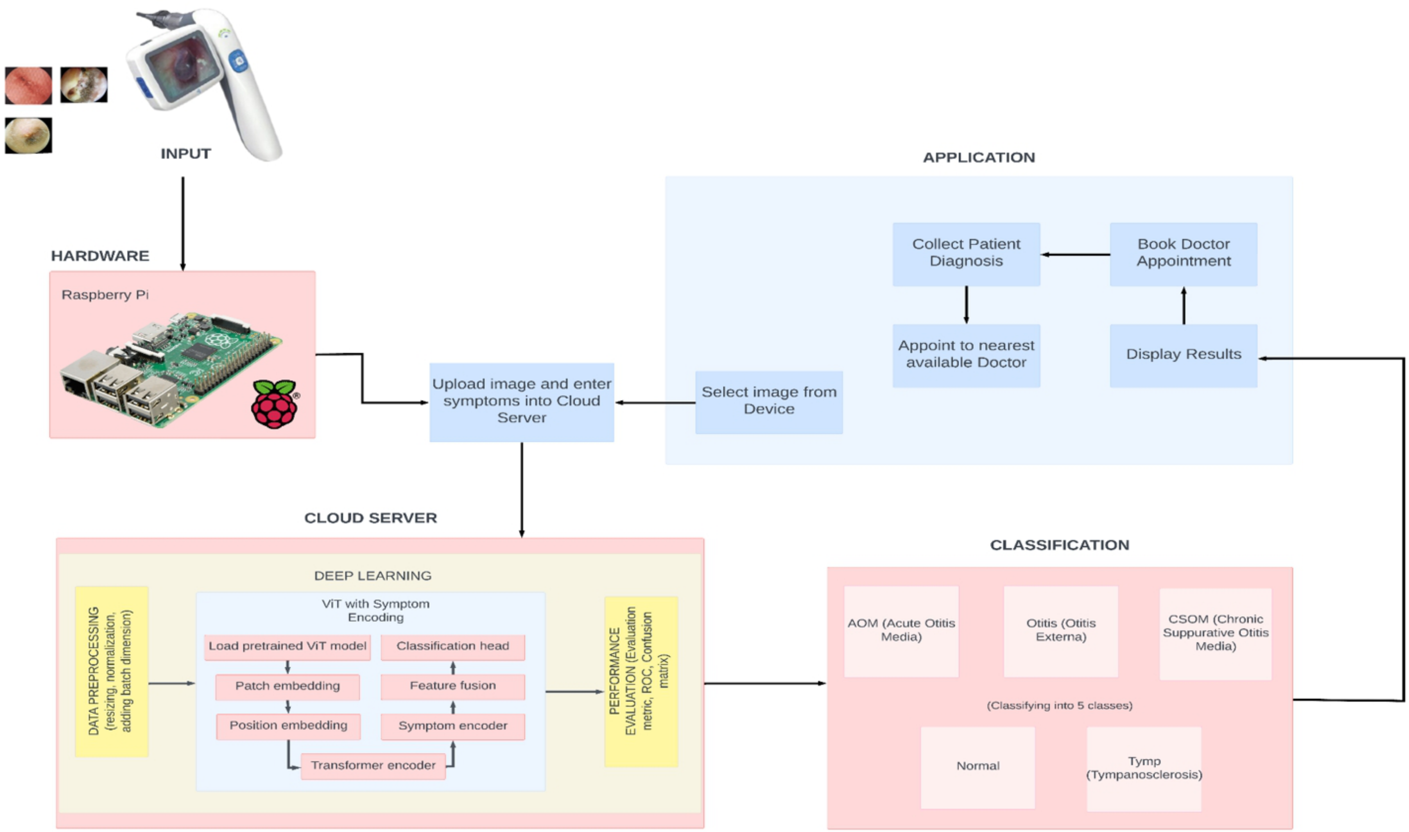

Fig. 1: Illustration of High Level Design

similar to the clinical diagnostic process, where symptoms and imaging findings are examined together [2].

The Kaggle "Otoscope Data" dataset, which includes otoscopic pictures of several middle ear infections, is used for training and evaluating the proposed approach. Preprocessing techniques including class balancing, data augmentation, and image enhancement are used to tackle dataset limitations. These techniques guarantee that the model is representative of many illness types and resilient to changes in image quality. Furthermore, this study enhanced the ViT architecture for computational efficiency, which makes it appropriate for deployment in environments with limited resources and hardware limitations.

Based on the experimental results, the proposed framework demonstrates superior performance compared to traditional CNN-based techniques regarding classification accuracy and robustness. The approach not only enhances diagnostic precision but also effectively merges automated image analysis with clinical practice by integrating visual data and symptomatology. This research provides a scalable and reliable resource for healthcare professionals, particularly in underserved regions. the project is a remarkable progress in the field of medical diagnostics.

II. Related Work

There are a number of limitations associated with the deep learning research currently being done on middle ear infection diagnosis, including dataset bias, challenges with data annotation, model generalization, interpretability issues, privacy and ethical concerns, and practical factors.

Crowson et al. introduced a methodology using deep learning models to diagnose middle ear infections. It achieved high diagnostic accuracy with a two-way predictive accuracy exceeding 90% and a three-way classification accuracy exceeding 88% [3]. However, the study faces limitations such as dataset biases that impacts the generalizability of the model to diverse populations. This causes challenges in implementing the methodology in real-world clinical settings. The lack of interpretability in deep learning models with low accuracy hinders trust among clinicians.

Al Rahim Habib et al. introduced a methodology for diagnosing ear diseases using deep learning techniques with otoscopic images. They trained convolutional neural networks such as DenseNet-161, ResNet-50, and Vision Transformer on three distinct cohorts of otoscopic images to evaluate both internal and external model performance. The internal validation showed high accuracy, but when applied to external cohorts, the models demonstrated a significant drop in performance [4].

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Cao et al. conducted a meta-analysis to evaluate the sensitivity and specificity of machine learning in diagnosing middle ear disorders using tympanic membrane images. Their findings revealed that deep learning-based models outperformed traditional machine learning algorithms [5]. However, the study's limitations includes biases in the datasets used, limited generalizability to diverse clinical settings, and reliance on high-quality tympanic membrane images for accurate predictions. Also, the black-box nature of CNN models limit clinical adoption due to challenges in interpretability and adoption.

Dahye Song et al. conducted a systematic review on the application of AI technologies in diagnosing middle ear diseases using image-based methods. They observed that combining segmentation with classification improved accuracy, with AI models outperforming manual diagnoses by medical specialists, achieving average diagnostic accuracies of 90.8%. However, limitations included the scarcity of data, inapplicability to diverse real-world cases, and issues with image quality, particularly in pediatric cases [6].

By utilizing a Vision Transformer integrated with feature fusion and symptom encoding, the proposed approach overcomes these limitations. This comprehensive methodology provides a dependable and effective way to diagnose middle ear infections while also improving diagnostic accuracy and bridging the gap between automated image analysis and actual clinical practice.

III. Proposed Methodology

The suggested method classifies five classes— Otitis Externa, CSOM, AOM, Tympanosclerosis, and Normal case. It does this by utilizing a Vision Transformer architecture coupled with symptom encoding in a process called feature fusion. Conventional CNN-based techniques mostly focus on local feature extraction. Whilst the Vision Transformer efficiently captures both local and global patterns by processing otoscopic pictures as a series of patches (Fig. 2).

Symptom encoding converts patient-reported symptoms into feature vectors. These are then integrated with image-derived features to provide comprehensive diagnosis of each case, enhancing model accuracy. Preprocessing methods like image enhancement, data augmentation, and class balancing were applied to ensure high-quality input data. The system was trained on the Kaggle 'Otoscope Data' dataset using optimized hyperparameters.

The infection diagnosis is achieved by combining the prediction result of the Vision Transformer (ViT) from the patient's otoscope image input. Also, the symptoms that the patient experiences are taken as additional inputs. These symptoms are encoded to complement the visual data. The final result is a combination of both of these to give an enhanced disease diagnosis.

This model addresses the limitations of existing approaches by utilizing the strengths of Vision Transformers (ViTs) [7] and symptom encoding. It gives us a reliable, accurate, and clinically applicable solution for the automated diagnosis of middle ear infections. [8] [9].

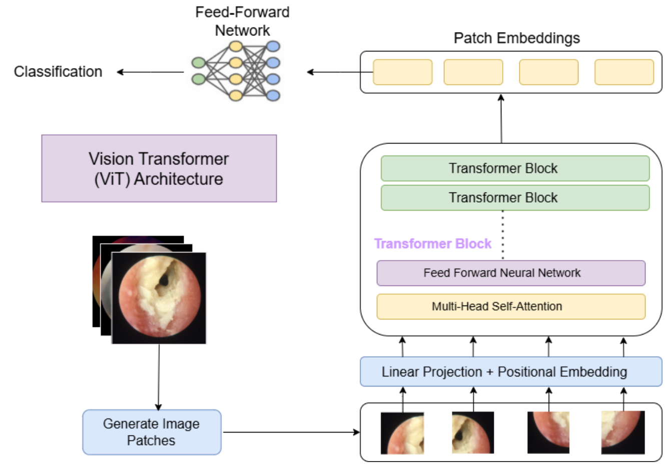

Fig. 2: Overview of proposed framework

IV. Dataset Selection

This study uses the 'Otoscope Data' dataset from Kaggle, which contains otoscopic images categorized into five classes. The five classes are Acute Otitis Media (AOM), Tympanosclerosis, Chronic Suppurative Otitis Media (CSOM), Otitis Externa, and normal ear conditions (Fig. 3). The dataset includes a variety of image resolutions, lighting conditions, and angles. It gives real-world clinical scenarios when live otoscopic images are captured using an otoscope. However, there are challenges like class imbalance, noise, and variations in image quality. They require careful preprocessing to improve the effectiveness of model training and evaluation [10] [11] [12].

Several preprocessing steps were applied to prepare the dataset for training. First, images were resized to meet the input requirements of the Vision Transformer (ViT) model. Data augmentation techniques including random rotations, flipping, and brightness adjustments were used to enhance dataset diversity and improve model generalization. To solve the issue of class imbalance, oversampling was applied to underrepresented categories. Weighted loss functions were also used during training. Additionally, normalization was performed to scale pixel values, to help in better model convergence. Low-quality or irrelevant images were removed during an initial exploratory analysis, to improve dataset reliability. These preprocessing measures optimized data quality. Therefore, the various data preprocessing steps allows the model to learn the patterns and achieve strong diagnostic performance.

V. Model Selection

The Vision Transformer (ViT) architecture, on which the model for this approach is built, performs exceptionally well in image classification tasks. Even if one is working with varying datasets and multi-modal inputs, the robust ViT architecture makes it easy to classify, which is predicting the correct diagnosis here. Convolutional Neural Networks (CNNs) use

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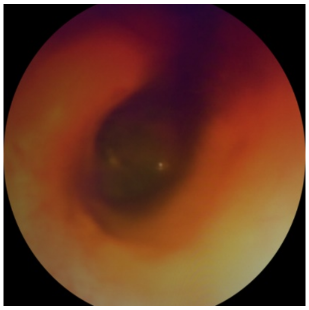
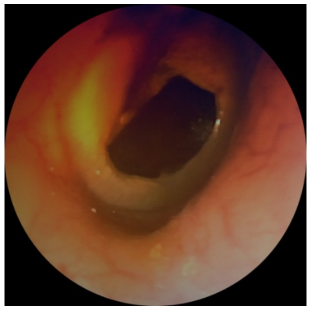
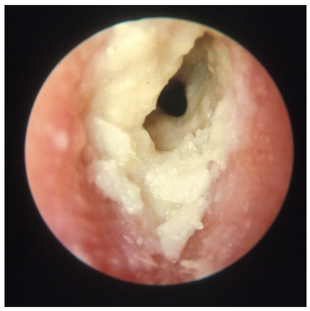


(a) Acute Otitis Media (AOM)



(b) Chronic Suppurative Otitis Media (CSOM)

(c) Tympanoskleros



(d) Otitis Externa

Fig. 3: Sample class images from the Otoscope Data dataset

convolutional operations to extract hierarchical spatial information as shown in previous studies [13] [14] [15]. Compared to that, the Vision Transformer uses attention mechanisms to process images through transformer blocks, treating them as a series of fixed-size patches. This method makes it possible for the model to recognize contextual links and long-range dependencies between remote areas of the image. This makes the ViT model very useful for medical imaging jobs where minute details are essential for a precise diagnosis [16] [17].

Otoscopic images, which are high-resolution pictures of the ear canal and eardrum used to diagnose middle ear infections, make up the model's input data. Initially, the images must be preprocessed to a uniform resolution of 224x224 pixels, normalized the pixel values to a range of [0, 1], and augmented (using rotation, flipping, scaling, and color jittering, among other techniques) as part of the data preparation process. By following these preprocessing processes, the photos are ensured to be high-quality, consistent, and sufficiently diversified to avoid overfitting during training.

The Vision Transformer model splits each image into non-overlapping patches, measuring 16 by 16 pixels. After the images have been preprocessed, the transformer model then flattens each patch into a vector and treats the series of patch vectors as tokens, similar to how words are handled in tasks involving natural language processing. This yields 14x14 patches (224 / 16 = 14) for a 224x224 pixel image, for a total of 196 patches per image.

The ViT's capacity to process images as sequences while maintaining the spatial links between patches is largely due to its use of positional embeddings. These embeddings are generated using a predetermined formula which provide information about the relative locations of patches within the image. The patch vectors are modified to include the positional encodings before being processed through the transformer encoder blocks.

The main component of Vision Transformer model are the various transformer encoder blocks. The ViT model which is used in this study, comprises 12 of such blocks in total in each of which there are two important building blocks, which are MHSA (Multi-Head Self-Attention) and FFNs (Feed Forward Neural networks). Here, the token sequence serves as image patches instead of text sentences. The MHSA allows patches to capture dependencies between distant regions of the image. Firstly, attention weights are calculated that decides how much attention each patch needs to pay to each other patch. Secondly, a weighted sum of the patch representations is outputted. The final layers of the transformer model aggregate this information and apply normalization before classification. MHSA helps parallelize the attention process across multiple heads. Position-wise FFN network consisting of two fully connected layers is present. A ReLU activation function is present in between the two layers. The output produced after the self-attention is passed through it. The introduction of ReLU enables the model to learn complex feature representations.

After this process, the model incorporates symptom information supplied with the images after visual features are extracted using ViT. Clinical metadata such as the patient's reported symptoms (pain, fever, or hearing loss) are represented as categorical variables. Each of the symptoms that the patient might be experiencing is transformed using one-hot encoding. The presence is represented as a binary vector. These feature vectors and included in the input data. Following their passage through the ViT blocks, these symptom vectors are concatenated with the feature vectors taken from the otoscopic pictures. Before being sent to the last classification layer, the combined feature vectors go through additional processing in a fully connected layer that combines visual and symptom data. The Vision Transformer outputs high dimensional features, while symptom encoding outputs the one-hot encoded vectors which are lower in dimension. Therefore, dimensionality reduction techniques are employed to negate the imbalance between the visual and clinical data. The fully connected layers learn how to associate the various symptom combinations with the visual patterns present in the infected otoscopic image, and this leads to enhanced disease diagnosis for middle ear infections.

Softmax function produces the probability distribution over the five possible classes: Acute Otitis Media (AOM), Tympanosclerosis, Chronic Suppurative Otitis Media (CSOM), Otitis Externa, and Normal (Fig. 3). This softmax function is what makes up the last classification layer. Based on the otoscopic image and the patient's clinical symptoms, the most likely condition is represented by the class with the highest probability. This is chosen as the model's projected diagnosis. This method allows the model to accept multi-modal input

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data, enhancing diagnostic accuracy by combining symptom-based knowledge (symptom encoding) with visual information (otoscope image).

To handle possible class imbalance in the training data, the model is trained by optimizing the parameters to minimize the loss function. A combination of cross-entropy loss and a weighted loss function is used to minimize the loss function. Gradient descent and backpropagation are used in the optimization process. This allows the model to iteratively adjust its parameters to lower prediction errors. To make sure to avoid effective convergence and overfitting, and improving accuracy, a variety of hyperparameters are adjusted: learning rate, batch size, and number of training epochs. Data augmentation techniques such as rotation, horizontal flip and contrast adjustments were employed. Additionally, AdamW optimizer was used to enhance model convergence and to help prevent overfitting. Accuracy, precision, recall, and F1-score are the validation measures used to track the model's performance during training, and have used them to further evaluate the model for generalization.

To improve the model's precision, focal loss technique was employed so that the bias of the model towards the majority classes can be negated by reducing their weights. Under-represented classes like Tympanosclerosis are assigned more weights and are focused on more. Additionally, weighted loss function is also used to ensure even learning across all classes. Performance of the model was improved by regularization techniques such as dropout and L2 weight decay.

Once the training is done, a separate test set obtained after the train-test data split is used to test the model. This testing dataset determines the model's performance on unknown data. The final model demonstrated its reliability in diagnosing middle ear infections with strong F1-scores in every class and an accuracy of 95.8%. Therefore this research shows how well the ViT model can identify minute details in otoscopic images, and obtain the correct diagnosis for the patient, particularly when paired with contextual symptom information.

In conclusion, the study's Vision Transformer model provides a very successful way to identify middle ear infections. Otoscope images coupled with symptom encoding helps increase the reliability and accuracy of the model. The ViT model is able to extract features that are essential for medical image analysis by segmenting images into patches and using self-attention mechanisms. Also, the model's diagnosis accuracy is improved by combining the otoscope image data with clinical symptom information during feature fusion This multi-modal approach enables it to generate well-informed predictions. This makes it a useful tool for helping medical practitioners diagnose middle ear disorders. With feature processing, weighted loss balancing and integration of focal loss, the model has managed to yield high results and provide high accuracy diagnosis combining otoscope image and patient symptoms.

VI. Experimental Results

The confusion matrix shows us a detailed analysis of the model's performance by comparing predicted and true labels. The number of true positives, true negatives, false positives and false negatives are necessary for calculating performance measures like accuracy, precision, recall, and F1-score (Fig. 4).

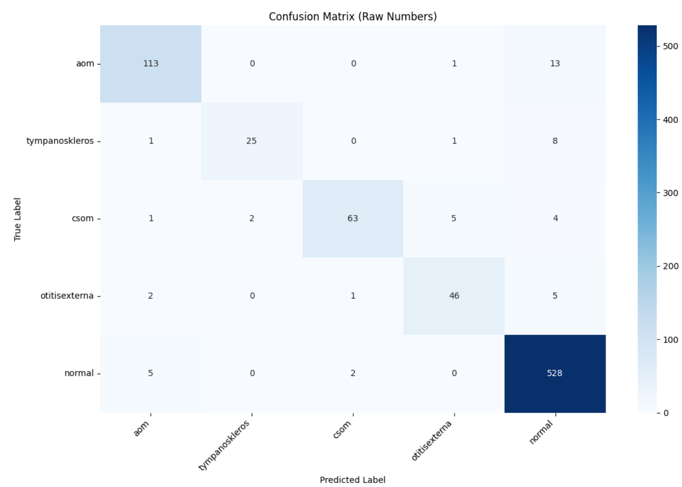

Fig. 4: Confusion matrix showing the model's performance

To evaluate the performance of the Vision Transformer (ViT) model in each of the five classes (Acute Otitis Media (AOM), Tympanosclerosis, Chronic Suppurative Otitis Media (CSOM), Otitis Externa, and Normal), this research has used precision, recall and F1-score. 96.6% of the cases classified as AOM were true positives, given by the model's precision of 0.966 for AOM. A comparatively high recall of 0.890 means that the model accurately identified 89.0% of real AOM events. AOM's F1-score of 0.926 shows us of the model's efficacy for this class, and helping to strike a balance between precision and recall. Likewise, the model maintains a high precision of 0.966 for Tympanosclerosis, assuring nearly all forecasts for this class are correct. The model can identify 80.0% of the real occurrences of Tympanosclerosis given by its recall score of 0.800. The model's reasonable performance for this circumstance is confirmed by the F1-score of 0.875 (Table I).

TABLE I: Model Evaluation Results

Class	Precision	Recall	F1-score
AOM	0.966	0.890	0.926
Tympanosklerosis	0.966	0.800	0.875
CSOM	0.986	0.907	0.944
Otitis Externa	0.912	0.963	0.937
Normal	0.957	0.991	0.973

For CSOM, the model performs exceptionally well with a precision of 0.986, which means that nearly all cases identified as CSOM are correct. The recall, at 90.7%, indicates that the model successfully detects the majority of actual CSOM cases. The model has highly reliable results with an F1-score of 0.944

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and there is a balance between precision and recall. When it comes to Otitis Externa, the model maintains a precision of 0.912 and a recall of 0.963. This tells us that it accurately captures most real cases of the condition, though with a slight dip in precision. The F1-score of 0.937 further confirms its strong diagnostic capability. For the Normal class, the model achieves an impressive precision of 0.957 and a recall of 0.991. This shows that it is highly effective at both correctly identifying normal cases, and distinguishing them from infected ones. The F1-score of 0.973 highlights the overall accuracy in classifying normal ear conditions. With an overall accuracy of 95.8% and a weighted F1-score of 0.957, the model displays remarkable consistency and accuracy across all categories. Thus, the model shows potential for practical applications in medical diagnosis, and to perform better diagnosis of the above mentioned middle ear diseases (Table I).

TABLE II: Model Evaluation Results - Overall Metrics

Metric	Value
Overall Precision	0.958
Macro avg F1-score	0.931
Weighted avg F1-score	0.957

The ROC curve for the model's performance is depicted in Figure 6. It illustrates the performance of the model across different thresholds. The curve demonstrates the trade-off between the True Positive Rate (TPR) and False Positive Rate (FPR), with the Area Under the Curve (AUC) for the model (Fig. 5)

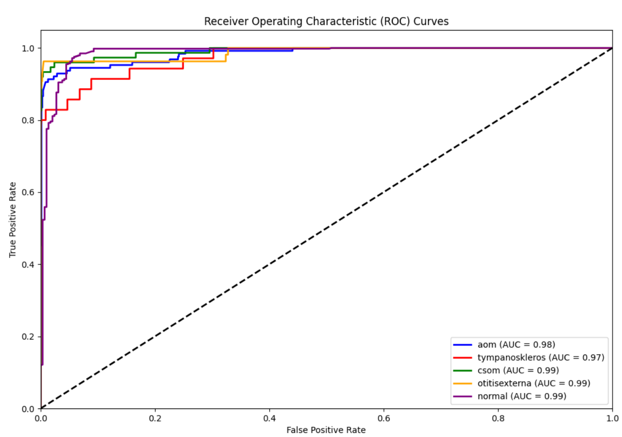

Fig. 5: Receiver Operating Characteristic (ROC) curve illustrating the performance of the model

VII. Conclusion

In conclusion, the proposed Vision Transformer (ViT) model with symptom encoding and feature fusion has performed exceptionally in classifying middle ear infections such as Acute Otitis Media, Tympanosclerosis, Chronic Suppurative Otitis Media, Otitis Externa, and Normal. Using the Kaggle 'Otoscope Data' dataset, the model achieved great precision, recall, and F1-scores for each class, demonstrating its robustness and reliability in medical diagnosis. The incorporation of symptom encoding in ViT improves the model's ability to analyze otoscopy pictures efficiently, resulting in accurate and rapid diagnosis, and provides higher performance than a standalone Vision Transformer model. The model's accuracy of 95.8% and weighted F1-score of 0.957 highlights its efficacy for clinical applications. This work demonstrates the ability of deep learning approaches for medical image processing, and also adds to the creation of AI-powered tools employing deep learning transformer model to help healthcare workers diagnose ear infections, ultimately improving patient outcomes and accelerating the diagnosis process. Future work can look into more model refinements as well as the addition of new symptoms and data sources to improve the system's capabilities.

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